JAVA PROJECT

REPORT

Course: Programming in Java (CSE2006)

Project Title: Technical Library & Resource Manager

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Submitted to: Dr Nilesh Kunhare

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1. The Problem & Motivation

I built **BYOP** because I was drowning in my own digital clutter. While juggling AI/ML studies and college event planning, I realized I had 19 research papers and dozens of Java tutorials lost in a black hole of folders and bookmarks.

I needed a tool that did more than just store files; I needed to track my comprehension. I wanted to see, at a glance, what I actually understood and what I still needed to learn. BYOP is my personal solution to the pre-semester-end scramble.

2. What the Application Does

I went with a **Java Console interface** because when I'm in the zone, I want speed—no distractions, just logic. Here's how it actually helps me stay organized:

- **Tidying up the mess:** It lets me file my papers under the right categories (AI, ML, or Java) as soon as I download them.
- **Knowing where I stand:** I added a status tracker so I can finally see which papers are actually "Read" and which ones are still on my "To-Do" list.
- **No more digging:** Instead of scrolling through folders, I can just type a keyword like "Ensemble" and the app pulls up the relevant research immediately.
- **It remembers everything:** Thanks to Java File I/O, all my updates are written to a text file. It's simple, reliable, and ready for me the next morning.

3.How I Structured the Code

I'll admit—at first, things were getting a bit messy. To fix that, I refactored the app into a **Layered Architecture**. I built a ResearchPaper class (using **Encapsulation**) to handle the data, and a LibrarySystem to handle the heavy lifting. One of my favorite "aha!" moments was switching to **pipes (|)** for my text file. Since academic titles are full of commas, a normal CSV would have corrupted my data immediately. Using pipes was a simple, "bulletproof" fix.

4.Putting Theory into Practice

I wanted to see if I could actually use what we learned in class to solve a personal problem:

- **OOP & Collections:** I used classes for every resource and an ArrayList so I could add as many papers as I needed without the app breaking.
- **File I/O:** I used BufferedReader and BufferedWriter to make sure my progress actually saved to my hard drive.
- **Crash-Proofing:** I added **Exception Handling** so that if I accidentally hit the wrong key while tired at 2:00 AM, the app just gives me an error message instead of quitting entirely.

5.Challenges & Growth

Building this tool taught me as much about project management as it did about Java:

- **Debugging Precision:** Encountering case-sensitivity issues with file naming reinforced the importance of naming conventions and attention to detail in backend development.
- **Version Control Strategy:** I moved beyond simple code storage to using GitHub as a storytelling tool. By committing the initial framework before the final logic, I successfully documented the iterative development process.
- **Scope Management:** Facing a tight end-of-March deadline, I practiced "Minimum Viable Product" (MVP) thinking. I pivoted from a web-based UI to a robust Console Application to ensure I delivered a high-quality, fully functional project for evaluation on time.

6. Conclusion

Building BYOP was a turning point for me. It helped me realize that Java isn't just a language for classroom exercises—it's a powerful tool for solving real-world data problems. I walked away from this project with more than just a working application to track my research; I gained hands-on experience in professional GitHub management and software architecture. This project proved that the best way to learn a language is to use it to solve your own challenges.

GitHub Link: https://github.com/shriyap0107/Java-Technical-Library